

陈泓森·Unity 脚本·复赛专业计划

你的核心价值

你是让 Cat Life“活起来”的人。Blender 做的模型在你手里变成可以交互的 App。

提交关联

傅钧烨资产 → 你搭建 Unity 场景 + 写脚本 → 严辰乐打包 APK → 傅钧漪录视频

Week 1 (5/26-6/1) : 工程搭建 + 场景导入

Unity 工程

Unity 2021.3 LTS / 2022.3 LTS

Platform: Android

Project: CatLife

```
Assets/  
  _Scenes/MainScene.unity  
  _Scripts/  
    Core/          # 状态机、数据管理  
    Cat/           # 猫咪行为控制  
    UI/            # UI管理器  
    LLM/           # 大模型调用 (对接吴若琪的LLMClient)  
  _Art/            # 傅钧烨提供  
    Models/        # FBX导入  
    Materials/      # 材质  
    Animations/     # 动画Controller  
  _Prefabs/  
  _Resources/
```

场景搭建 (对接傅钧烨的 FBX)

1. 导入猫主角 + 建筑 + 道具
2. 设置地面 (Plane + 草绿色材质)
3. 灯光 : 1 Directional Light (暖色) + 环境光
4. 相机 : Perspective , 3/4 俯视角度
5. Canvas : Screen Space Overlay (UI 层)

猫咪基础脚本

```
// CatController.cs  
public class CatController : MonoBehaviour {  
    public Animator animator;  
  
    public void PlayIdle() => animator.SetTrigger("Idle");  
    public void PlayApproach() => animator.SetTrigger("Approach");  
    public void PlayRest() => animator.SetTrigger("Rest");  
    public void PlayCelebrate() => animator.SetTrigger("Celebrate");  
  
    // 简单随机动作 (普通状态下)  
    IEnumerator RandomIdleActions() {  
        while (state == CatState.Normal) {  
            yield return new WaitForSeconds(Random.Range(3f, 8f));  
            // 随机小动作 : 伸懒腰、眨眼、转头  
        }  
    }  
}
```

```
}  
}
```

Week 2 (6/2-6/8) : 状态机 + 动画

四状态状态机

```
public enum CatState { Normal, Transition, Focus, Reward }  
  
public class StateMachine : MonoBehaviour {  
    public CatState currentState = CatState.Normal;  
  
    public void SwitchState(CatState newState) {  
        currentState = newState;  
        switch (newState) {  
            case CatState.Normal:    OnEnterNormal();    break;  
            case CatState.Transition: OnEnterTransition(); break;  
            case CatState.Focus:     OnEnterFocus();     break;  
            case CatState.Reward:    OnEnterReward();    break;  
        }  
    }  
}
```

双入口专注

```
// 入口 1: 手动点击 " 开始专注 "  
public void OnStartFocusClicked() {  
    SwitchState(CatState.Focus);  
    UIManager.Instance.ShowFocusUI();  
}  
  
// 入口 2: 自动识别 (行为检测)  
void Update() {  
    if (currentState == CatState.Normal) {  
        float activity = behaviorTracker.GetActivityRate(30f);  
        if (activity < transitionThreshold) {  
            SwitchState(CatState.Transition);  
        }  
    }  
}
```

猫咪动画状态对应

状态	动画	循环
Normal	Idle (含随机小动作)	
Transition	Approach (走近 + 注视)	
Focus	Rest (安静趴着)	
Reward	Celebrate (跳起 + 星星)	1 次

如果没时间做骨骼动画, 用 **Transform** 动画替代: - Idle: 小幅上下浮动 - Approach: 向镜头移动 + 放大 - Rest: 缩小 + 变半透明 - Celebrate: 旋转 + 弹跳

Week 3 (6/9-6/15) : 轻锁定 UI+ 行为识别

轻锁定 UI

```
public class FocusUIManager : MonoBehaviour {
    public GameObject[] hideInFocus; // 专注时隐藏的按钮
    public GameObject swipeUpHint;   // "↑ 上滑退出"

    public void EnterFocus() {
        foreach (var obj in hideInFocus)
            obj.SetActive(false);
        swipeUpHint.SetActive(true);
        StartCoroutine(FocusTimer());
    }

    public void ExitFocus() {
        foreach (var obj in hideInFocus)
            obj.SetActive(true);
        swipeUpHint.SetActive(false);
    }

    // 上滑检测
    void Update() {
        if (Input.touchCount > 0) {
            Touch t = Input.GetTouch(0);
            if (t.phase == TouchPhase.Ended && t.deltaPosition.y > 100) {
                ExitFocus();
            }
        }
    }
}
```

行为数据采集

```
public class BehaviorTracker : MonoBehaviour {
    public int clickCount30s;
    public int swipeCount30s;
    public float idleSeconds;

    void Update() {
        if (Input.GetMouseButtonDown(0)) clickCount30s++;
        // Touch 滑动
        if (Input.touchCount > 0 && Input.GetTouch(0).phase == TouchPhase.Moved)
            swipeCount30s++;
        if (!Input.anyKey && Input.touchCount == 0)
            idleSeconds += Time.deltaTime;
    }

    public float GetActivityRate(float window) {
        return (clickCount30s + swipeCount30s) / window;
    }

    // 每 30 秒重置计数器
    IEnumerator ResetWindow() {
        while (true) {
            yield return new WaitForSeconds(30f);
            clickCount30s = 0; swipeCount30s = 0;
        }
    }
}
```

```

    }
}

```

对接吴若琪的 LLMClient

```

public class SmartFocusAnalyzer : MonoBehaviour {
    public LLMClient llmClient; // 吴若琪提供

    public void AnalyzeWithLLM() {
        var data = new BehaviorData {
            clickCount = behaviorTracker.clickCount30s,
            swipeCount = behaviorTracker.swipeCount30s,
            idleSeconds = behaviorTracker.idleSeconds
        };
        llmClient.AnalyzeFocusState(data, (result) => {
            // 解析 JSON, 决定是否切换状态
            // {"state": "focus", "score": 0.87}
        });
    }
}

```

Week 4 (6/16-6/23) : 联调 + 录屏素材

联调

- 与傅钧烨：确认所有资产在 Unity 中正确
- 与吴若琪：LLMClient 集成测试
- 与严辰乐：Unity 导出给 Android 集成
- 与傅钧漪：提供录屏用的场景配置

录屏素材准备

- 场景配置为最佳展示角度
- 预设演示流程：开屏 → 领养 → 活跃 → 点专注 → 安静 → 庆祝
- 隐藏调试 UI
- 确保帧率流畅

脚本交付清单

```

unity_scripts/
  StateMachine.cs      # 四状态机
  CatController.cs     # 猫咪行为
  BehaviorTracker.cs   # 行为采集
  FocusUIManager.cs   # 轻锁定UI
  SmartFocusAnalyzer.cs # LLM集成
  DataManager.cs       # 本地存储
  UIManager.cs        # 页面管理

```